Passive Impulsive Cube Balancing

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This exercise of Classical Dynamics is inspired by the Cubli[1], a cubic robot that is able to balance itself on its edges and corners via reaction wheels. While Cubli accomplishes this via an active controller, I explore the possibility of accomplishing this passively with a single impulsive action.

Note that this problem is a slight departure to the magnetorquer-assisted reaction wheel desaturation problem I originally proposed. The reason for this departure is due to complexities in magnetorquer mechanics that unnecessarily complicate the problem and are out-of-scope of the course.

I. Problem Statement

Consider a uniform cube of mass M and side length a resting on a frictionless surface. The cube has two reaction wheel embedded within. Each reaction wheel can be modeled as a uniform thin disc with mass m and radius $r < \frac{a}{2}$. Assume the masses m of the wheels neither contribute to the mass M of the cube nor impact its uniformity. The wheels are oriented such that the normal-vector of wheel 1 is orthoganol to the front face of the cube and the normal-vectof of wheel 2 is forms a 45^o angle with the top and right faces of the cube. The center-of-mass of the two reaction wheels coincide with the center-of-mass of the uniform cube. Finally, the wheels are fixed with respect to the cube and spin on frictionless bearings.

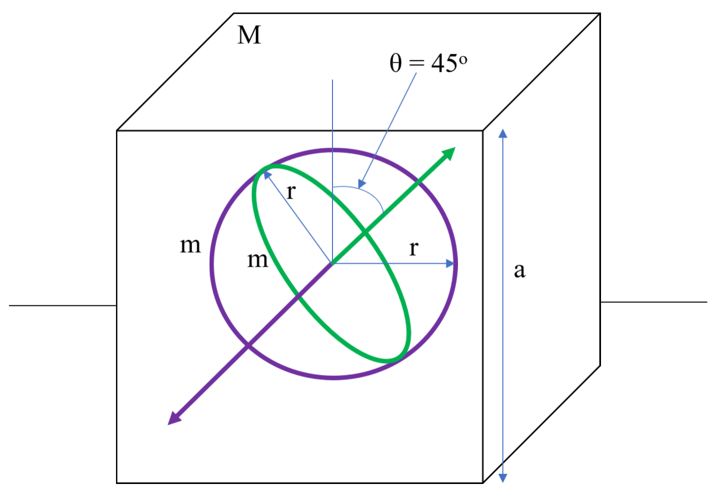

Fig. 1 Cube with embedded reaction wheels. Purple is wheel 1. Green is wheel 2.

Wheel 1 is initially spinning at an angular velocity ω_1 and wheel 2 is initially at rest. Wheel 1 is suddenly braked to a stop (assume no thermal loss) and as a result, the cube stands up and perfectly balances on its edge. Wheel 2 is then suddenly accelerated to ω_2 and the cube ends up perfectly balancing itself on one of it's corners. This sequence of actions is demonstrated in Figure 2.

Derive the set of differential equations governing the dynamics each action that allow you to solve for ω_1 and ω_2. (Actually solving the system is unnecessary).

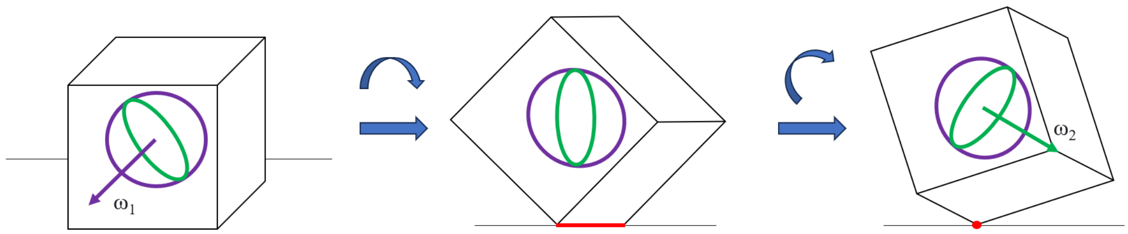

Fig. 2 Visual for how the cube progresses from initial to final state. Red indicates balancing edge & vertex respectively. Top arrows roughly indicate direction of angular displacement between successive states as an intuitive aid.

II. Technical Approach

A. Problem Setup

1) Define the System: The system is defined as the cube + 2 reaction wheel system.

2) Define the Reference Frame: The inertial reference frame will be fixed with respect to the ground. The origin of the reference frame will coincide with the **initial** center-of-mass of the system. The selection of this reference frame is designed such that the position vector of the body frame relative to the inertial frame $\vec{r}_{B/S}$ is initially $\vec{0}$. The body reference frame will be fixed with respect to the cube. The origin of the body frame will always coincide with the center-of-mass of the system and will rotate with the system. This was chosen as the body frame because it is easiest to find the inertia tensor of the system $\{I_{sys}\}$ about the origin of the body frame, since all 3 bodies in the system share the origin of the body frame as a center-of-mass.

3) Define the Coordinate System: For all rigid bodies in space, the coordinate system is a 6-tuple of values (x, y, z position coordinates and θ, ϕ, ψ rotation coordinates). It is evident that the dynamics of this problem only require two Euler Angles as rotation coordinates θ and ϕ. Less immediately evident is the fact that throughout the horizon of both transformations, there are only moments and z-direction forces being imparted on the system (recall the cube rests on a frictionless surface). As a result, $\dot{x} = 0$ and $\dot{y} = 0$ by conservation of linear momentum. Qualitatively, this implies the center-of-mass of the system only sees linear motion in the z-component of the inertial frame. z is defined as the third coordinate[Appendix A][Appendix B].

B. Solution: Transformation 1 (Edge Balancing)

Refer to figure 3 for the free-body-diagram of this problem. This problem has 2 generalized coordinates θ and z. This problem has 1 nontrivial holonomic constraint:

$$z = \frac{\sqrt{2}}{2} a \sin\left(\theta + \frac{\pi}{4}\right)$$

It follows that this problem only has 1 degree-of-freedom (2 generalized coordinates - 1 holonomic constraint).

$\vec{p}_x$ and $\vec{p}_y$ are both conserved because there is no external force acting on the system in the x, y component.
$\vec{H}_{sys}$ is not conserved since external moments are being applied through wheel-deceleration and gravity.
Total energy of the system is not conserved since the sudden deceleration of wheel 1 to be in rotational lock with the cube is an inelastic collision. However, energy is conserved post-impulse.

First, find the x-component principal moment of inertia of the system with respect to the x-axis of the body

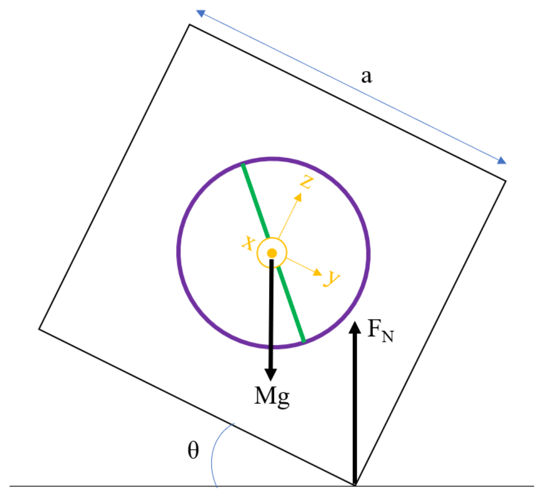

Fig. 3 Front View Free Body Diagram of the system post-impulse for transformation 1.

frame $I_{xx/B}$. For the rest of this problem, the inertial subscript xx/B is implied. The inertia of the system is denoted I_{sys}

$$I_{cube} = \frac{1}{12}M(a^2 + a^2) \qquad \text{moment of inertia of a cube rotating about its principal axis}$$
$$I_{w1} = \frac{1}{2}mr^2 \qquad \text{moment of inertia of a thin disc spinning on its central axis}$$
$$I_{w2} = \frac{1}{4}mr^2 \qquad \text{moment of inertia of a thin disc spinning on its vertical axis}$$
$$I_{sys} = I_{cube} + I_{w1} + I_{w2}$$

Next, solve for the angular velocity of the system right after wheel 1 is brought to a stop $\dot{\theta}_i$. To do so, apply conservation of angular momentum.

$$\begin{aligned} I_{w1}\Delta\omega_{w1} &= I_{sys}\Delta\dot{\theta} \qquad \text{Conservation of } \vec{H}_{sys} \\ I_{w1}(-\dot{\theta}_i - \omega_{w1}) &= I_{sys}(-\dot{\theta}_i) \\ \dot{\theta}_i &= \frac{I_{w1}\omega_{w1}}{I_{sys} - I_{w1}} \end{aligned}$$

First, try using Force and Moment balance to solve for the dynamics of the system. Force balance with the holonomic constraint applied yields the following:

$$F_N - (M + 2m)g = (M + 2m)\ddot{z} = (M + 2m)\frac{\sqrt{2}}{2}a\left(\ddot{\theta}\cos(\theta + \pi/4) - \dot{\theta}^2\sin(\theta + \pi/4)\right) \tag{1}$$

Moment balance yields the following:

$$F_N * \frac{\sqrt{2}}{2}a\cos(\theta + \pi/4) = -I_{sys}\ddot{\theta} \tag{2}$$

Equipped with equations (1) and (2), it is possible to solve for the differential equation governing θ.

$$F_N = -\frac{I_{sys}}{\frac{\sqrt{2}}{2}a\cos(\theta + \pi/4)}\ddot{\theta}$$
$$-\frac{I_{sys}}{\frac{\sqrt{2}}{2}a\cos(\theta + \pi/4)}\ddot{\theta} - (M + 2m)g = (M + 2m)\frac{\sqrt{2}}{2}a\left(\ddot{\theta}\cos(\theta + \pi/4) - \dot{\theta}^2\sin(\theta + \pi/4)\right)$$
$$I_{sys}\ddot{\theta} + \frac{\sqrt{2}}{2}ag(M + 2m)\cos(\theta + \pi/4) + \frac{1}{2}(M + 2m)a^2\ddot{\theta}\cos^2(\theta + \pi/4) - \frac{1}{4}(M + 2m)a^2\dot{\theta}^2\cos(2\theta) = 0$$

Next, try using a Lagrangian approach to solving the system.

$$T = T_{rot} + T_{trans}$$
$$T_{rot} = \frac{1}{2} I_{sys} \dot{\theta}^2$$
$$T_{trans} = \frac{1}{2}(M + 2m)\dot{z}^2 = \frac{1}{4} a^2 \dot{\theta}^2 (M + 2m) \cos^2(\theta + \pi/4)$$
$$V = \frac{\sqrt{2}}{2}(M + 2m) g a \sin(\theta + \pi/4)$$

Set up the Lagrangian

$$\mathcal{L} = T - V = \frac{1}{2} I_{sys} \dot{\theta}^2 + \frac{1}{2}(M + 2m)\dot{z}^2 = \frac{1}{4} a^2 \dot{\theta}^2 (M + 2m) \cos^2(\theta + \pi/4) - \frac{\sqrt{2}}{2}(M + 2m) g a \sin(\theta + \pi/4) \tag{3}$$

Solve the Lagrangian for general coordinate θ

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = I_{sys}\dot{\theta} + \frac{1}{2}\dot{\theta} a^2 (M + 2m) \cos^2(\theta + \pi/4)$$
$$\frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}) = I_{sys}\ddot{\theta} + \frac{1}{2}\ddot{\theta} a^2 (M + 2m) \cos^2(\theta + \pi/4) - \frac{1}{2}\dot{\theta}^2 a^2 (M + 2m) \cos(2\theta)$$

$$\frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$
$$I_{sys}\ddot{\theta} + \frac{1}{2}\ddot{\theta} a^2 (M + 2m) \cos^2(\theta + \pi/4) - \frac{1}{4}\dot{\theta}^2 a^2 (M + 2m) \cos(2\theta) + \frac{\sqrt{2}}{2}(M + 2m) g a \cos(\theta + \pi/4) = 0$$

The solution obtained using the Lagrangian approach is consistent with that taken from the Force & Moment Balance approach!
Thus we have found the differential equation governing θ

$$\boxed{I_{sys}\ddot{\theta} + \frac{1}{2}\ddot{\theta} a^2 (M + 2m) \cos^2(\theta + \pi/4) - \frac{1}{4}\dot{\theta}^2 a^2 (M + 2m) \cos(2\theta) + \frac{\sqrt{2}}{2}(M + 2m) g a \cos(\theta + \pi/4) = 0} \tag{4}$$

subject to the following boundary conditions

1) $\theta_i = 0$, $\dot{\theta}_i = \frac{I_{w1}\omega_{w1}}{I_{sys} - I_{w1}}$
2) When $\theta = \pi/4$, $\dot{\theta} = \ddot{\theta} = 0$.

Equation 3 is a nonlinear differential equation, and will hence require numerical methods to solve. However, we can see that given the 1st set of boundary conditions, it is possible to solve the system in terms of ω_{w1}. Then, applying the second set of boundary conditions, ω_{w1} can be solved for.

C. Solution: Transformation 2 (Vertex Balancing)

As a preface, this problem follows the same structure and procedure as the previous problem because they are similar in nature. The key differences to be aware of are the difference in the moment of inertia of the system I_{sys} and the fact that wheel 2 itself is now a rotating body whilst also rotating with the system.

Refer to figure 4 for the free-body-diagram of this problem.
This problem has 2 generalized coordinates ϕ and z.
This problem has 1 nontrivial holonomic constraint:

$$z = \frac{\sqrt{3}}{2} a \sin\left(\phi + \arccos\left(\frac{\sqrt{3}}{3}\right)\right)$$

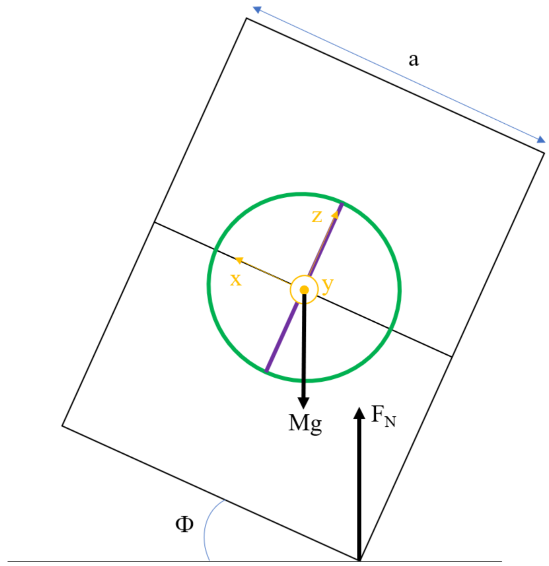

Fig. 4 Right View Free Body Diagram of the system post-impulse for transformation 2.

It follows that this problem only has 1 degree-of-freedom (2 generalized coordinates - 1 holonomic constraint).

$\vec{p}_x$ and $\vec{p}_y$ are both conserved because there is no external force acting on the system in the x, y component.
$\vec{H}_{sys}$ is not conserved since external moments are being applied through wheel-deceleration and gravity.
Total energy of the system is not conserved since the sudden acceleration of wheel 2 requires energy input. However, energy is conserved post-impulse.

First, find the moment of inertia of the system about the y-axis of a $\pi/4$ rotated body frame B' denoted $I_{yy/B'}$. For the rest of this problem, the inertial subscript yy/B' is implied. In this case, the cube is no longer rotating about a principal axis while the two wheels still are. Thus, the moments of inertia of the wheels remain simple, albeit swapped from Transformation 1.

$$I_{w1} = \frac{1}{4}mr^2$$
$$I_{w2} = \frac{1}{2}mr^2$$

To find the moment of inertia of the cube about a non-principal axis, we must use the principal inertia tensor of the cube $\{I_B\}$ and apply the proper rotation tensor $\overleftrightarrow{R}$. In this case, the new body frame is rotated $\pi/4$ from the original body frame about the x-axis. Using the rotation tensor appropriate for this, we find $\{I_{B'}\}$.

$$\begin{aligned}
\{I_{B'}\} &= \overleftrightarrow{R}^{-1}\{I_B\}\overleftrightarrow{R} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\pi/4) & -\sin(\pi/4) \\ 0 & \sin(\pi/4) & \cos(\pi/4) \end{bmatrix}^{-1} \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\pi/4) & -\sin(\pi/4) \\ 0 & \sin(\pi/4) & \cos(\pi/4) \end{bmatrix} \\
&= \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & \frac{1}{2}(I_{zz}+I_{yy}) & \frac{1}{2}(I_{zz}-I_{yy}) \\ 0 & \frac{1}{2}(I_{zz}-I_{yy}) & \frac{1}{2}(I_{zz}+I_{yy}) \end{bmatrix}
\end{aligned}$$

Thus, taking only the y-principal component of $\{I_{B'}\}$, we obtain

$$I_{cube} = \frac{1}{2}(I_{zz}+I_{yy}) = \frac{1}{12}M(a^2+a^2)$$

Notice that the moment of inertia is identical to those about a principal axis. This is expected from a geometry that is perfectly symmetric about its principal axes ($I_{xx} = I_{yy} = I_{zz}$).

Next, we once again solve for the angular velocity of the system $\dot{\phi}_i$ right after wheel 2 is accelerated to ω_{w2} by applying conservation of angular momentum.

$$I_{w2}(\omega_{w2} - \dot{\phi}) = I_{sys}(-\dot{\phi})$$
$$\dot{\phi} = \frac{I_{w2}\omega_{w2}}{I_{w2} - I_{sys}}$$

Before continuing with solving the complete Force & Moment balance equations again, it is important to notice that if you substitute ϕ for θ and $(\phi + 0.955)$ (where $0.955 \approx \arccos\left(\frac{\sqrt{3}}{3}\right)$ which is just a constant offset-angle) for $(\theta + \phi/4)$ this problem has the exact same setup as Transformation 1, with the only difference being that wheel 2 has its own constant angular velocity throughout. This fact only impacts the setup of the moment-balance equation in the balance approach and only impacts T_{rot} in the Lagrangian approach. As a result, we can isolate these two components, solve for them, and see what impact they will have on the final system dynamics.

First we analyze the effects of ω_{w2} on the moment-balance equation.

$$\begin{aligned}
\vec{M} &= \frac{d\vec{H}}{dt} \Rightarrow M = \dot{H} \\
H &= -(I_{w1} + I_{cube})\dot{\phi} + I_{w2}(\omega_{w2} - \dot{\phi}) \\
\dot{H} &= -(I_{w1} + I_{w2} + I_{cube})\ddot{\phi} + I_{w2}\dot{\omega_{w2}} & \dot{\omega_{w2}} = 0 \\
\dot{H} &= M = -I_{sys}\ddot{\phi}
\end{aligned}$$

Substituting the proper term in for M yields the following equation.

$$F_N * \frac{\sqrt{3}}{2} a \cos(\phi + 0.955) = -I_{sys}\ddot{\phi} \tag{5}$$

Compare equation (5) with equation (2). Besides a few changes in scalar constant coefficients and swapping ϕ for θ, the moment-balance equation is unaffected by a constant ω_{w2}! Next we proceed to analyze the effect of ω_{w2} on T_{rot} and subsequently the Lagrangian $\mathcal{L}$.

$$\begin{aligned}
T_{rot} &= \frac{1}{2}(I_{w1} + I_{cube})\dot{\phi}^2 + \frac{1}{2}I_{w2}(\omega_{w2} - \dot{\phi})^2 \\
&= \frac{1}{2}(I_{w1} + I_{cube})\dot{\phi}^2 + \frac{1}{2}I_{w2}(\omega_{w2}^2 + 2\dot{\phi}\omega_{w2} + \dot{\phi}^2) \\
&= \frac{1}{2}I_{sys}\dot{\phi}^2 + \frac{1}{2}I_{w2}(\omega_{w2}^2 + 2\dot{\phi}\omega_{w2}) \\
\frac{\partial T_{rot}}{\partial \dot{\phi}} &= I_{sys}\dot{\phi} + I_{w2}\omega_{w2} \\
\frac{d}{dt}(\frac{\partial T_{rot}}{\partial \dot{\phi}}) &= I_{sys}\ddot{\phi}
\end{aligned}$$

Once again, we see that besides the ϕ for θ swap, ω_{w2} has no impact on the solved Lagrangian!
Without needing to resolve the entire system, we can find the differential equation governing ϕ by form-matching our result from Transformation 1 [equation (4)].

$$\boxed{I_{sys}\ddot{\phi} + \frac{1}{2}\ddot{\phi}a^2(M + 2m)\cos^2(\phi + 0.955) - \frac{1}{4}\dot{\phi}^2 a^2(M + 2m)\sin(2\phi + 1.91) + \frac{\sqrt{3}}{2}(M + 2m)ga\cos(\phi + 0.955) = 0} \tag{6}$$

subject to the following constraints:

1) $\phi_i = 0$, $\dot{\phi}_i = \frac{I_{w2}\omega_{w2}}{I_{w2} - I_{sys}}$
2) When $\phi = \pi/2 - 0.955$, $\dot{\phi} = \ddot{\phi} = 0$.

Once again, this equation is a non-linear differential equation that should be solved numerically. ω_{w2} can subsequently be solved for using the boundary conditions.

III. Summary

A. Qualitative Interpretation

Listed a few critical qualitative takeaways from the Passive Impulsive Cube Balancing problem.

1) The dynamics of this problem are non-linear. This is consistent with the classic inverted pendulum problem, a cousin of this balancing cube problem.
2) The absence of friction between the surface and the system, the x-y position of the center-of-mass of the system does not change due to conservation of momentum in the x-y plane.
3) The moment of inertia of a cube about any axis through its center-of-mass is always equal to the principle moments of inertia of the cube. This is a by-product of a cube being a symmetric object where $I_x x = I_y y = I_z z$. This implies the moment of inertia of the system is consistent between Transformations 1 and 2.
4) The presence of a frictionless reaction wheel with constant angular velocity relative to the system body frame does not impact the system dynamics and produces the same differential equation as if the reaction wheel was not spinning.

B. Comparison of Approaches

For this problem, both the Force-Balance and Lagrangian solutions were reasonably approaches in terms of difficulty. However the Lagrangian solution is slightly easier. Force-balance requires solving a system of 2 equations due to F_N being an additional unknown. In the Lagrangian formulation, F_N is not considered as a general force due to it arising from the holonomic constraints.

C. Changes to Problem Definition

This problem assumed a frictionless surface. However, the real Cubli operates on a surface that ideally has a high coefficient of friction in order to effectively turn the point-of-contact between the cube and the ground into a hinge joint (for edge-balancing) or a ball joint (for vertex-balancing) that is stationary with respect to the inertial reference frame. Friction introduces an additional nontrivial force in the x-y plane, causing conservation of linear momentum to break down as well as adding general forces to the Lagrangian interpretation.

This problem also starts with an awkward initial condition where wheel 1 is spinning while wheel 2 is stationary. This choice of initial conditions is deliberately designed to zero-out the total angular momentum of the system in the intermediate, edge-balanced state. If total angular momentum was nonzero in the intermediate state, an external orthogonal impulse on the system will cause precession.

Lastly, in practice, in order to balance on a vertex, three reaction wheels are required in order to maintain 3-degrees of control freedom.

IV. Appendix

A.

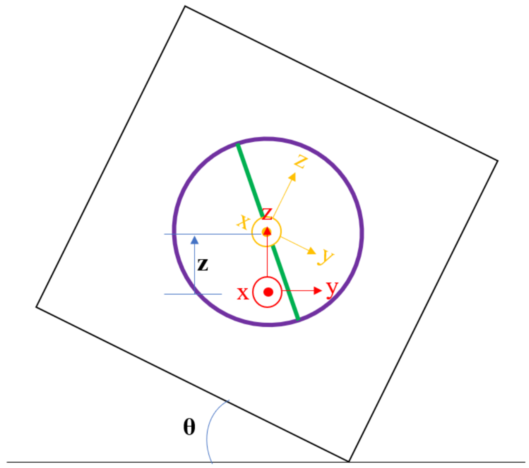

Fig. 5 Front view of cube during transformation 1 showing coordinates θ and z. Red is the inertial frame, Orange is the body frame.

B.

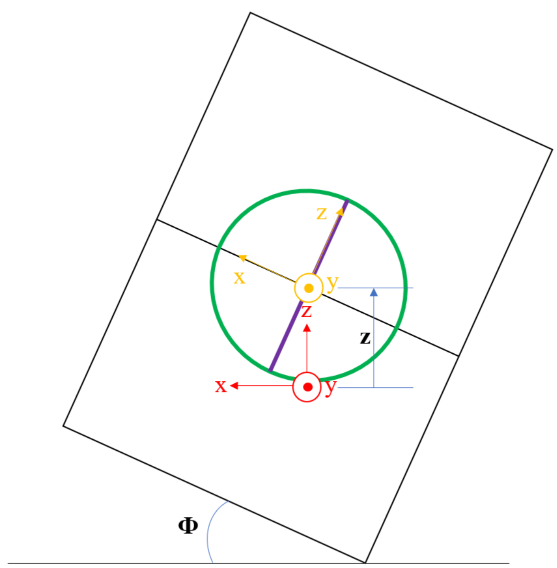

Fig. 6 Right view of cube during transformation 2 showing coordinates ϕ and z. Red is the inertial frame, Orange is the body frame.

References

[1] M. Gajamohan, M. Muehlebach, T. Widmer and R. D'Andrea, "The Cubli: A reaction wheel based 3D inverted pendulum," 2013 European Control Conference (ECC), Zurich, Switzerland, 2013, pp. 268-274, doi: 10.23919/ECC.2013.6669562.